

# **SONiX 32-Bit Cortex-M0 Micro-Controller**

## ***DMA Memory Example Code***

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AMENDENT HISTORY

| Version | Date       | Description       |
|---------|------------|-------------------|
| 1.0     | 2024/11/27 | 1. First version. |

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# Table of Content

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|                                      |          |
|--------------------------------------|----------|
| AMENDMENT HISTORY .....              | 2        |
| <b>1 OVERVIEW.....</b>               | <b>4</b> |
| <b>2 FEATURE .....</b>               | <b>4</b> |
| <b>3 HW .....</b>                    | <b>4</b> |
| <b>4 EXAMPLE CODE.....</b>           | <b>5</b> |
| 4.1 INITIALIZATION .....             | 5        |
| 4.2 DMA FLASH TO SRAM TRANSFER ..... | 6        |
| 4.3 DMA SRAM TO SRAM TRANSFER.....   | 7        |
| <b>5 RESULT .....</b>                | <b>8</b> |
| 5.1 DMA FLASH TO SRAM TRANSFER ..... | 8        |
| 5.2 DMA SRAM TO SRAM TRANSFER.....   | 9        |

# 1 OVERVIEW

This document provides our customers the C program examples of DMA Memory-to-memory transfer mode.

# 2 FEATURE

- DMA FLASH to SRAM transfer example.
- DMA SRAM to SRAM transfer example.
- Use ARM-compiler V6.12

# 3 HW

- The starter kit board of vendor desired MCU.

## 4 EXAMPLE CODE

### 4.1 Initialization

Initial DMA function. User can modify the settings of DMA in these function. The setting of DMA should refer to the table in the datasheet.

```
97 // Init DMA
98 DMAMtoMInit();
```

```
161 void DMAMtoMInit(void)
162 {
163     DMA_InitSt stDMACH_Init;
164
165     // Set DMA CH0 as M(SRAM/FLASH) to M(SRAM/FLASH) mode
166     // Source : Memory, Increment, 8bit, no request, Burst 1
167     stDMACH_Init.b_SrcAddrCtrl = DMA_SRCAD_CTL_INC;
168     stDMACH_Init.b_SrcMode     = DMA_SRC_MEMORY;
169     stDMACH_Init.b_SrcWidth    = DMA_SRC_WIDTH_8BIT;
170     stDMACH_Init.b_SrcReqSel   = DMA_SRCRS_NONE;
171     stDMACH_Init.b_BurstSize   = DMA_BURST_1;
172     // Destination : Memory, Fixed, 8bit, no request
173     stDMACH_Init.b_DstAddrCtrl = DMA_DSTAD_CTL_INC;
174     stDMACH_Init.b_DstMode     = DMA_DST_MEMORY;
175     stDMACH_Init.b_DstWidth    = DMA_DST_WIDTH_8BIT;
176     stDMACH_Init.b_DstReqSel   = DMA_DSTRS_NONE;
177     // Channel : LV0, Enable TC/ABT interrupt
178     stDMACH_Init.b_Priority    = DMA_CHPRI_LV0;
179     stDMACH_Init.b_IntTCEn     = DMA_INT_TC_MSK_DIS;
180     stDMACH_Init.b_IntABTEn    = DMA_INT_ABT_MSK_DIS;
181
182     // Initial DMA
183     DMA_Init(&stDMACH_Init, eDMA_CH0);
184 }
```

## 4.2 DMA FLASH to SRAM transfer

The source data are 256 bytes of random data in FLASH. The destination address is SRAM.

```

45 // Random Flash data
46 const uint8_t b_aryFlashSrcData[DMA_TEST_SIZE] =
47 {
48     0x59, 0x79, 0x69, 0xfd, 0x9d, 0xe3, 0xf0, 0xef, 0x2d, 0xed, 0xb4, 0xb4, 0x05, 0x8b, 0xff, 0x08,
49     0x19, 0xe5, 0x47, 0x2f, 0x2b, 0x33, 0x49, 0x30, 0x82, 0x04, 0x62, 0x9a, 0xdd, 0xec, 0xa3, 0x3b,
50     0xc5, 0xf9, 0x85, 0x38, 0x54, 0x8f, 0x5d, 0xa0, 0x0f, 0x19, 0x07, 0x0b, 0xdd, 0xb0, 0xea, 0x33,
51     0xb9, 0x77, 0x25, 0x10, 0xc6, 0x84, 0x6c, 0xec, 0x32, 0xb5, 0x94, 0x70, 0x4d, 0x68, 0xa3, 0x33,
52     0x81, 0x4c, 0x33, 0x20, 0x3b, 0x10, 0xf6, 0x78, 0x90, 0x79, 0x13, 0xaf, 0xe5, 0xe0, 0x83, 0x02,
53     0xe9, 0xad, 0xec, 0x63, 0x5f, 0xc5, 0x6e, 0xd1, 0xaa, 0x6b, 0xc5, 0xaf, 0xf9, 0x8c, 0x84, 0x8f,
54     0x27, 0xf3, 0xb3, 0xe4, 0xb7, 0xf0, 0xc2, 0x14, 0x7e, 0x37, 0xa1, 0xc9, 0xa0, 0xf3, 0xed, 0x2e,
55     0x50, 0x17, 0xa7, 0xbe, 0x65, 0x98, 0xeb, 0xa2, 0x6a, 0x7d, 0xdf, 0x06, 0x69, 0xe1, 0x5e, 0x17,
56     0xd7, 0x31, 0xdc, 0x00, 0xb3, 0xa3, 0x7c, 0x20, 0x3b, 0x3d, 0x91, 0xad, 0x95, 0x3c, 0x57, 0xbf,
57     0x18, 0x39, 0x82, 0xed, 0x90, 0x10, 0xec, 0x4d, 0xaf, 0x1d, 0x5c, 0xd0, 0x84, 0x67, 0xe6, 0x90,
58     0xff, 0x9c, 0xa9, 0xf3, 0xa0, 0xc6, 0xaa, 0x39, 0x44, 0xa5, 0x79, 0x01, 0xab, 0x35, 0x38, 0x6d,
59     0xe5, 0x54, 0xf0, 0xc5, 0x43, 0xd6, 0x44, 0xb5, 0x1d, 0xc5, 0x50, 0x7b, 0x3b, 0xfd, 0xa2, 0xfb,
60     0x60, 0xae, 0xa4, 0x3b, 0xa5, 0xe5, 0x64, 0xc3, 0xbc, 0x02, 0x14, 0xff, 0xe2, 0xd5, 0xe1, 0x44,
61     0x65, 0x94, 0x04, 0x0b, 0x5d, 0xc9, 0xbd, 0xb3, 0xbd, 0x4d, 0x10, 0xff, 0x54, 0x01, 0x80, 0x2b,
62     0xb5, 0x25, 0xf3, 0x04, 0xb0, 0x97, 0x01, 0x09, 0xdd, 0xf9, 0x5b, 0x11, 0xeb, 0x31, 0x0a, 0x4a,
63     0x60, 0x10, 0x20, 0xca, 0x07, 0x6c, 0x39, 0x8b, 0x72, 0xb2, 0x79, 0xa4, 0x4d, 0x2a, 0xb9, 0x3d
64 };

```

After DMA is enabled, DMA starts transmitting from FLASH address to SRAM address. The DMA TC flag and interrupt will be set when all data transfers are completed.

```

109 /* Example 1: DMA Flash to SRAM Start*/
110
111 // Start DMA transfer
112 DMAFlashToSramStart();
113
114 // Wait for DMA TC interrupt flag
115 while(stDMA0IntFlag[eDMA_CH0].Flag.bits.TC != 1);
116
117 // Check SRAM data = FLASH data
118 for(i = 0; i < DMA_TEST_SIZE; i++)
119 {
120     if(b_aryFlashSrcData[i] != b_arySramDstData[i])
121         while(1); //Fail
122 }
123
124 /* Example 1: DMA Flash to SRAM End*/

```

### 4.3 DMA SRAM to SRAM transfer

The source data are 256 bytes of data in SRAM. The destination address is SRAM.

```
100 // Init data buffer
101 for(i = 0; i<DMA_TEST_SIZE ;i++)
102 {
103     // Set source data = 0x00, 0x01, ... ,0xFF
104     b_arySramSrcData[i] = i & 0xFF;
105     // Set destination data = 0x00
106     b_arySramDstData[i] = 0x00;
107 }
```

After DMA is enabled, DMA starts transmitting from source SRAM address to destination SRAM address. The DMA TC flag and interrupt will be set when all data transfers are completed.

```
133 /* Example 2: DMA SRAM to SRAM Start*/
134
135 // Start DMA transfer
136 DMASramtoSramStart();
137
138 // Wait for DMA TC interrupt flag
139 while(stDMA0IntFlag[eDMA_CH0].Flag.bits.TC != 1);
140
141 // Check TX data = RX data
142 for(i = 0; i<DMA_TEST_SIZE ;i++)
143 {
144     if(b_arySramSrcData[i] != b_arySramDstData[i])
145         while(1); //Fail
146 }
147
148 /* Example 2: DMA SRAM to SRAM End*/
```

## 5 RESULT

The results can be observed through the Memory window.

### 5.1 DMA FLASH to SRAM transfer

The data in the destination SRAM will be written as the data in source FLASH.



| Memory 2                  |                                                 |
|---------------------------|-------------------------------------------------|
| Address: b_arySramDstData |                                                 |
| 0x2000010C:               | 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 |
| 0x2000011C:               | 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 |
| 0x2000012C:               | 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 |
| 0x2000013C:               | 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 |
| 0x2000014C:               | 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 |
| 0x2000015C:               | 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 |
| 0x2000016C:               | 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 |
| 0x2000017C:               | 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 |
| 0x2000018C:               | 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 |
| 0x2000019C:               | 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 |
| 0x200001AC:               | 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 |
| 0x200001BC:               | 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 |
| 0x200001CC:               | 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 |
| 0x200001DC:               | 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 |
| 0x200001EC:               | 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 |
| 0x200001FC:               | 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 |

  

| Memory 2                  |                                                 |
|---------------------------|-------------------------------------------------|
| Address: b_arySramDstData |                                                 |
| 0x2000010C:               | 59 79 69 FD 9D E3 F0 EF 2D ED B4 B4 05 8B FF 08 |
| 0x2000011C:               | 19 E5 47 2F 2B 33 49 30 82 04 62 9A DD EC A3 3B |
| 0x2000012C:               | C5 F9 85 38 54 8F 5D A0 0F 19 07 0B DD B0 EA 33 |
| 0x2000013C:               | B9 77 25 10 C6 84 6C EC 32 B5 94 70 4D 68 A3 33 |
| 0x2000014C:               | 81 4C 33 20 3B 10 F6 78 90 79 13 AF E5 E0 83 02 |
| 0x2000015C:               | E9 AD EC 63 5F C5 6E D1 AA 6B C5 AF F9 8C 84 8F |
| 0x2000016C:               | 27 F3 B3 E4 B7 F0 C2 14 7E 37 A1 C9 A0 F3 ED 2E |
| 0x2000017C:               | 50 17 A7 BE 65 98 EB A2 6A 7D DF 06 69 E1 5E 17 |
| 0x2000018C:               | D7 31 DC 00 B3 A3 7C 20 3B 3D 91 AD 95 3C 57 BF |
| 0x2000019C:               | 18 39 82 ED 90 10 EC 4D AF 1D 5C D0 84 67 E6 90 |
| 0x200001AC:               | FF 9C A9 F3 A0 C6 AA 39 44 A5 79 01 AB 35 38 6D |
| 0x200001BC:               | E5 54 F0 C5 43 D6 44 B5 1D C5 50 7B 3B FD A2 FB |
| 0x200001CC:               | 60 AE A4 3B A5 E5 64 C3 BC 02 14 FF E2 D5 E1 44 |
| 0x200001DC:               | 65 94 04 0B 5D C9 BD B3 BD 4D 10 FF 54 01 80 2B |
| 0x200001EC:               | B5 25 F3 04 B0 97 01 09 DD F9 5B 11 EB 31 0A 4A |
| 0x200001FC:               | 60 10 20 CA 07 6C 39 8B 72 B2 79 A4 4D 2A B9 3D |



## 5.2 DMA SRAM to SRAM transfer

The data in the destination SRAM will be written as the data in source SRAM.



| Memory 2    |                                                 |
|-------------|-------------------------------------------------|
| Address:    | b_arySramDstData                                |
| 0x2000010C: | 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 |
| 0x2000011C: | 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 |
| 0x2000012C: | 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 |
| 0x2000013C: | 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 |
| 0x2000014C: | 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 |
| 0x2000015C: | 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 |
| 0x2000016C: | 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 |
| 0x2000017C: | 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 |
| 0x2000018C: | 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 |
| 0x2000019C: | 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 |
| 0x200001AC: | 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 |
| 0x200001BC: | 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 |
| 0x200001CC: | 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 |
| 0x200001DC: | 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 |
| 0x200001EC: | 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 |
| 0x200001FC: | 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 |

  

| Memory 2    |                                                 |
|-------------|-------------------------------------------------|
| Address:    | b_arySramDstData                                |
| 0x2000010C: | 00 01 02 03 04 05 06 07 08 09 0A 0B 0C 0D 0E 0F |
| 0x2000011C: | 10 11 12 13 14 15 16 17 18 19 1A 1B 1C 1D 1E 1F |
| 0x2000012C: | 20 21 22 23 24 25 26 27 28 29 2A 2B 2C 2D 2E 2F |
| 0x2000013C: | 30 31 32 33 34 35 36 37 38 39 3A 3B 3C 3D 3E 3F |
| 0x2000014C: | 40 41 42 43 44 45 46 47 48 49 4A 4B 4C 4D 4E 4F |
| 0x2000015C: | 50 51 52 53 54 55 56 57 58 59 5A 5B 5C 5D 5E 5F |
| 0x2000016C: | 60 61 62 63 64 65 66 67 68 69 6A 6B 6C 6D 6E 6F |
| 0x2000017C: | 70 71 72 73 74 75 76 77 78 79 7A 7B 7C 7D 7E 7F |
| 0x2000018C: | 80 81 82 83 84 85 86 87 88 89 8A 8B 8C 8D 8E 8F |
| 0x2000019C: | 90 91 92 93 94 95 96 97 98 99 9A 9B 9C 9D 9E 9F |
| 0x200001AC: | A0 A1 A2 A3 A4 A5 A6 A7 A8 A9 AA AB AC AD AE AF |
| 0x200001BC: | B0 B1 B2 B3 B4 B5 B6 B7 B8 B9 BA BB BC BD BE BF |
| 0x200001CC: | C0 C1 C2 C3 C4 C5 C6 C7 C8 C9 CA CB CC CD CE CF |
| 0x200001DC: | D0 D1 D2 D3 D4 D5 D6 D7 D8 D9 DA DB DC DD DE DF |
| 0x200001EC: | E0 E1 E2 E3 E4 E5 E6 E7 E8 E9 EA EB EC ED EE EF |
| 0x200001FC: | F0 F1 F2 F3 F4 F5 F6 F7 F8 F9 FA FB FC FD FE FF |

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